ANTOINE RATOUCHNIAK https://antoinertk.github.io/

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WORK EXPERIENCE

Amazon *Edinburgh, United Kingdom*
Applied Scientist 05/2025 - Present

- Developed hyperbolic representation learning methods for job recommendations, improving downstream hiring model performance by 5-8% across multiple teams.
- Designed an LLM-as-a-Judge optimization approach using implicit gradients and built an automated pipeline that generated ~3,000 qualification assessment questions deployed on Amazon.jobs.
- First-authored two papers at the *Amazon Machine Learning Conference* on hyperbolic representation learning and implicit-gradient optimization of LLM judges (18.1% and 9.1% acceptance rates).

Meta Platforms, FAIR (Fundamental AI Research) *London, United Kingdom*
Research Scientist Intern 05/2024 - 11/2024

- Investigated self-supervised pre-training and cross-task transfer for neural decoding across image, speech, and language modalities under the supervision of Hubert J. Banville and Jean-Rémi King.
- Co-authored NeuralBench and NeuralSet, two open-source projects for NeuroAI benchmarking and machine learning.

New York University, **NYU Video Lab** *New York City, United States*
Visiting Researcher 03/2023 - 09/2026

- Developed machine learning methods to predict language-critical cortical regions from ECoG recordings, under the supervision of Yao Wang and Adeen Flinker.

PUBLICATIONS

NeuralBench: A Unifying Framework to Benchmark NeuroAI Models
H. Banville et al. (including *A. Ratouchniak*), 2026. arXiv

NeuralSet: A High-Performing Python Package for Neuro-AI
JR. King et al. (including *A. Ratouchniak*), 2026. arXiv

Prediction of Stimulation-Defined Eloquent Cortex Using Graph-Theoretical Connectivity from Electrocorticography During Presurgical Mapping
N. Emami et al. (including *A. Ratouchniak*), 2026. bioRxiv

Machine Learning-Based Prediction of Speech Arrest During Direct Cortical Stimulation Mapping
N. Emami et al. (including *A. Ratouchniak*), IEEE International Conference on Neural Engineering, 2025. arXiv

EDUCATION

ENS Paris-Saclay 2023 - 2024
M.Sc. in mathematics and artificial intelligence, MVA *Gif-sur-Yvette, France*
Coursework: convex optimization, probabilistic graphical models, optimal transport, time series, geometric data analysis, bayesian machine learning, kernel methods, algorithms for speech, mathematics for neurosciences

Paris-Saclay University 2021 - 2022
M.Sc. in mathematics and artificial intelligence *Orsay, France*
Coursework: probability, statistics, deep learning, data structures, graph theory, information theory, optimization

Gustave Eiffel University 2018 - 2021
B.Sc. in mathematics and computer science *Champs-sur-Marne, France*

ACADEMIC SERVICE

Reviewer – AAAI 2027

SKILLS

Languages French (native), English (fluent), Spanish (intermediate), Chinese (basic)

Programming/Tools Python (scikit-learn, PyTorch, Hugging Face), Java, Apache Spark, Git, AWS